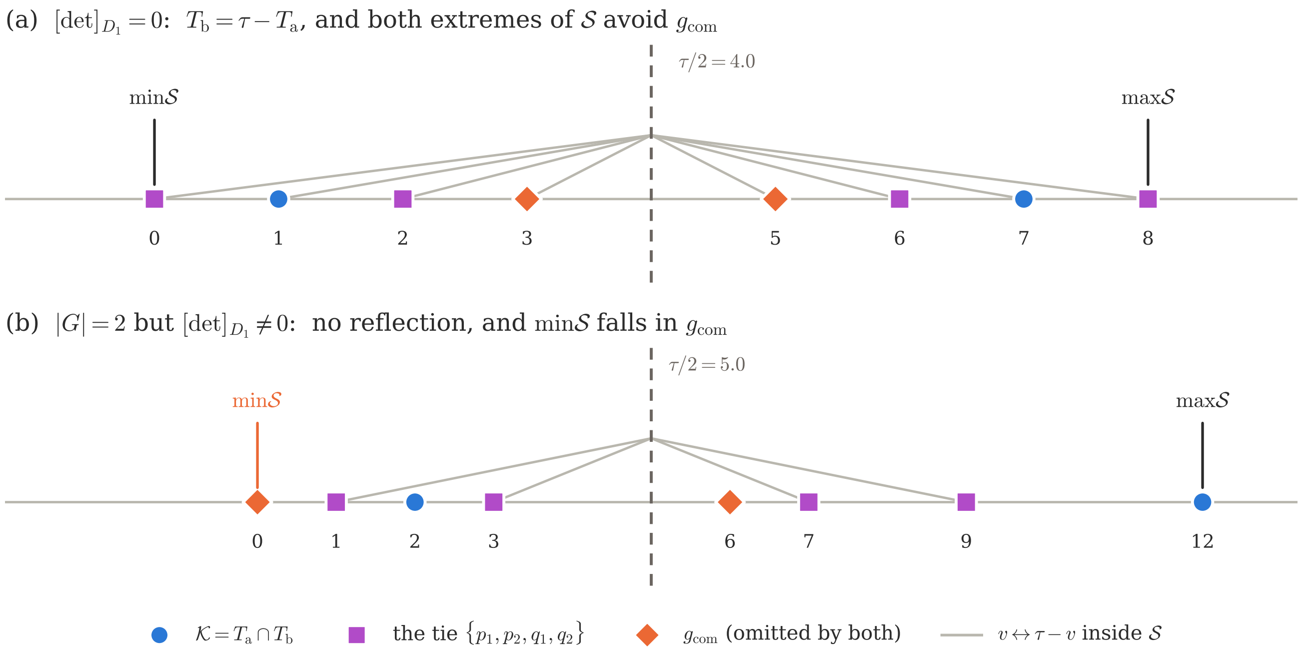

(a) $[\det]_{D_1} = 0$: $T_b = \tau - T_a$, and both extremes of $\mathcal{S}$ avoid g_{com}
$\tau/2 = 4.0$
$\min\mathcal{S}$
$\max\mathcal{S}$
0
1
2
3
5
6
7
8
(b) $|G| = 2$ but $[\det]_{D_1} \neq 0$: no reflection, and $\min\mathcal{S}$ falls in g_{com}
$\tau/2 = 5.0$
$\min\mathcal{S}$
$\max\mathcal{S}$
0
1
2
3
6
7
9
12
$\mathcal{K} = T_a \cap T_b$
the tie $\{p_1, p_2, q_1, q_2\}$
g_{com} (omitted by both)
$v \leftrightarrow \tau - v$ inside $\mathcal{S}$